Aptitude Preparation Notes

Quantitative aptitude, reasoning, and placement-test fundamentals

Percentages

- Percentage = (part ÷ whole) × 100. To find x% of y, calculate (x ÷ 100) × y.
- Percentage increase = (increase ÷ original value) × 100. Percentage decrease = (decrease ÷ original value) × 100.
- Example: 20% of 250 = 50. If 100 increases by 20% and then decreases by 20%, the result is 96, not 100.
- Practice: Find 15% of 640. A product costs ■800 and rises by 10%; find the new price. A value falls from 500 to 425; find the percentage decrease.

Profit, Loss, and Discount

- Profit = Selling Price – Cost Price. Loss = Cost Price – Selling Price.
- Profit percentage = (profit ÷ cost price) × 100. Loss percentage = (loss ÷ cost price) × 100.
- Discount = Marked Price – Selling Price. Discount percentage = (discount ÷ marked price) × 100.
- Practice: An item bought for ■600 is sold for ■750. Find profit percentage. A marked price of ■1,200 has a 15% discount; find selling price.

Ratio and Proportion

- A ratio compares quantities. If a:b = 2:3 and the total is 50, there are five parts, each worth 10; the quantities are 20 and 30.
- A proportion states that two ratios are equal, such as a/b = c/d.
- Practice: Divide ■840 in the ratio 3:4. If five notebooks cost ■150, find the cost of eight at the same rate.

Time and Work

- If a person completes a job in n days, their work rate is 1/n of the job per day.
- If A takes 6 days and B takes 3 days, their combined rate is 1/6 + 1/3 = 1/2 job per day, so together they take 2 days.
- Practice: A completes work in 10 days and B in 15 days. How long do they take together?

Speed, Distance, and Time

- Speed = Distance ÷ Time. Distance = Speed × Time. Time = Distance ÷ Speed.
- Keep units consistent. 1 km/h = 5/18 m/s and 1 m/s = 18/5 km/h.
- Practice: A car travels 180 km in 3 hours. Find average speed. Convert 72 km/h to m/s.

Averages and Reasoning

- Average = sum of observations ÷ number of observations. If the average of five values is 12, their sum is 60.
- Common reasoning topics include number series, analogies, coding-decoding, directions, blood relations, syllogisms, and arrangements.
- For number series, inspect differences, ratios, alternating patterns, squares, and cubes.
- Practice: Find the next number: 2, 6, 12, 20, 30, __. If you face north and turn right, which direction are you facing?

Verbal Ability

- Common topics include reading comprehension, sentence correction, subject-verb agreement, tenses, vocabulary, and para jumbles.
- Read the complete sentence before choosing an answer. Check grammatical agreement and the relationship between ideas.

• Practice: Choose the correct form: 'She (work/works) every day.'